

Zae Myung Kim

Fourth-year CS PhD candidate at University of Minnesota Twin Cities

My research centers on a “**meta-scaffolding paradigm**” for LLMs that injects metacognitive structure across training and inference. This paradigm (1) integrates **discourse-level meta-structures** into training to improve coherence and human-like long-form generation; (2) embeds **metacognitive feedback** into reinforcement learning to stabilize optimization, reduce reward hacking, and minimize manual prompt engineering; and (3) advances **test-time metacognition** for LLM agents, where a universal meta-operation applied recursively builds hierarchies of self-improving layers whose roles (e.g., remediation, routing, specialization) emerge from increased dimensional context, allowing agents to surpass their base performance through the heightened metacognition and insight accumulated across layers.

I have collaborated with researchers at Amazon, Salesforce, Grammarly, Naver, and Samsung.

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🏠 zaemyung.github.io

🐙 [GitHub](#)

🎓 [Google Scholar](#)

🌐 [LinkedIn](#)

🐦 [zaemyung](#)

Education

Sept 2022 – Mar 2027 (expected)	University of Minnesota Twin Cities PhD in Computer Science Thesis: <i>Meta-Scaffolding for Large Language Models</i> (tentative) Advisor: Dongyeop Kang	Minneapolis, MN
Sept 2014 – Aug 2016	Korea Advanced Institute of Science and Technology (KAIST) MS in Computer Science Thesis: <i>Classifying Travel-Related Intents in Textual Data Using Recurrent Neural Networks</i> Advisor: Ho-Jin Choi	Daejeon, Korea
Oct 2008 – Aug 2011	Imperial College London BEng in Computer Science Group Project: <i>Intention Recognition in Ambient Homes</i> Individual Project: <i>Computing Fictitious Plays in Graph-Based Games</i>	London, UK

Professional Employment

Sept – Dec 2026	Research Intern – Amazon AWS Developing a metric recommendation system for Amazon EC2 ASG using time-series and heterogeneous data streams.	Seattle, WA
Jun – Nov 2024	Research Intern – Amazon AGI Proposed and developed structural alignment of LLMs for long-form generation.	Bellevue, WA
May – Sept 2023	Research Intern – Salesforce AI Research Implemented a large-scale challenging benchmarking system for scientific-paper retrieval. Built a co-writing system with large language models for scientific writing.	Palo Alto, CA
Apr – Jul 2022	Research Intern – Grammarly Developed a human-in-the-loop iterative text-revision system. Trained span-based text-revision models with data augmentation.	Remote
Oct 2020 – Feb 2022	ML Researcher – Naver Labs Europe Studied language-specific representations in multilingual NMT. Conducted cross-lingual discourse analysis for document-level NMT.	Meylan, France

Mar 2017 – Sept 2020	ML Researcher – Papago, Naver Corp. Developed machine translation models for Thai–Korean and German–Korean pairs. Implemented tools for NMT: sentence-segmenter , corpus pre-processor, aligner, and crawler. Created website-translation service using NMT + heuristic attention. Ranked 1 st (doc-level) and 4 th (sent-level) at WMT20 quality estimation shared task.	Seongnam, Korea
Dec 2016 – Feb 2017	Invited Researcher – KEAI Lab, KAIST Built an adaptive vocabulary learning app using learner interest profiles. Developed a healthcare app that recommends Pokémon Go events by forecasting users’ physical activity patterns using recurrent neural network.	Daejeon, Korea
Sept – Nov 2016	Research Intern – Naver Labs Corp. Researched extending named entity recognition for Korean with hierarchical categories.	Seongnam, Korea
Apr 2014 – Jul 2016	Research Assistant – Samsung–KAIST Initiative [1 st Year] Intent classification for travel-service personalization. [2 nd Year] Modeled long-term activeness using RNNs for wellness prediction.	Daejeon, Korea
Dec 2013 – Sept 2014	Research Intern – KEAI Lab, KAIST Surveyed plan and intention recognition. Researched temporal information extraction for English and Korean.	Daejeon, Korea
Jan 2012 – Oct 2013	Military Service – Special Warfare Command, ROK Army Served as an army interpreter, aviation image analyst, and squad leader.	Incheon, Korea

Honors, Awards, and Accomplishments

2025	Doctoral Dissertation Fellowship (University of Minnesota)
2022	Best Paper Award (In2Writing workshop at ACL 2022)
2022	3M Science and Technology Fellowship (University of Minnesota)
2018	Runner-Up , Elice Data Science Challenge (3rd place out of 300 participants)
2016	Outstanding Teaching Assistant Award (2016 Elice Data Science Workshop & Challenge)
2016	Outstanding Graduate Teaching Assistant Award (KAIST)
2012	Runner-Up , Republic of Korea Army Oratorical Contest (ROK Army)
2011	Winning Team , Robotics Localization Challenge (Imperial College London)
2009	Winning Team , Deutsche Bank Prize for Topics in AI Group Project (Imperial College London)
2008	Bronze Medal , Royal College of Science Union Science Challenge (Imperial College London)

Publications

An asterisk (*) marks co-first authors.

2026	Metaⁿ: Recursive Self-Improvement through Emergent Depth Zae Myung Kim, Young-Jun Lee, Seungyeon Jwa, Dongyeop Kang <i>NeurIPS 2026.</i>
2026	Structure Liberates: How Constrained Sensemaking Produces More Novel Research Output James Mooney, Zae Myung Kim, Young-Jun Lee, Dongyeop Kang. <i>Under review.</i>
2026	The Amazing Agent Race: Strong Tool Users, Weak Navigators Zae Myung Kim, Dongseok Lee, Jaehyung Kim, Vipul Raheja, Dongyeop Kang. <i>Under review.</i>

- 2026 **Align to Structure: Aligning Large Language Models with Structural Information**
Zae Myung Kim, Anand Ramachandran, Farideh Tavazoei, Joo-Kyung Kim, Oleg Rokhlenko, Dongyeop Kang.
AAAI 2026.
- 2025 **Toward Evaluative Thinking: Meta Policy Optimization with Evolving Reward Models**
Zae Myung Kim, Chanwoo Park, Vipul Raheja, Dongyeop Kang.
Under review.
- 2025 **Anchor Aweigh! Sail for Optimal Unified Multi-Modal Representations**
Minoh Jeong*, Zae Myung Kim*, Min Namgung*, Dongyeop Kang, Yao-Yi Chiang, Alfred Hero.
Under review.
- 2024 **Threads of Subtlety: Detecting Machine-Generated Texts Through Discourse Motifs**
Zae Myung Kim, Kwang Hee Lee, Preston Zhu, Vipul Raheja, Dongyeop Kang.
ACL 2024.
- 2024 **Benchmarking Cognitive Biases in Large Language Models as Evaluators**
Ryan Koo, Minhwa Lee, Vipul Raheja, Jong Inn Park, Zae Myung Kim, Dongyeop Kang.
Findings of ACL 2024.
- 2024 **Under the Surface: Tracking the Artifactuality of LLM-Generated Data**
Debarati Das*, Karin de Langis*, Anna Martin-Boyle*, Jaehyung Kim*, Minhwa Lee*, Zae Myung Kim*, Shirley Anugrah Hayati, Risako Owan, Bin Hu, Ritik Sachin Parkar, Ryan Koo, Jong Inn Park, Aahan Tyagi, Libby Ferland, Sanjali Roy, Vincent Liu, Dongyeop Kang.
Under review.
- 2022 **Improving Iterative Text Revision by Learning Where To Edit From Other Revision Tasks**
Zae Myung Kim, Wanyu Du, Vipul Raheja, Dhruv Kumar, Dongyeop Kang.
EMNLP 2022.
- 2022 **Read, Revise, Repeat: A System Demonstration for Human-in-the-loop Iterative Text Revision [Best Paper Awarded]**
Wanyu Du*, Zae Myung Kim*, Vipul Raheja, Dhruv Kumar, Dongyeop Kang.
In2Writing at ACL 2022.
- 2022 **Understanding Iterative Revision from Human-Written Text**
Wanyu Du, Vipul Raheja, Dhruv Kumar, Zae Myung Kim, Melissa Lopez, Dongyeop Kang.
ACL 2022.
- 2021 **Visualizing Cross-Lingual Discourse Relations in Multilingual TED Corpora**
Zae Myung Kim, Vassilina Nikoulina, Dongyeop Kang, Didier Schwab, Laurent Besacier.
CODI 2021 at EMNLP 2021.
- 2021 **Do Multilingual Neural Machine Translation Models Contain Language Pair Specific Attention Heads?**
Zae Myung Kim, Laurent Besacier, Vassilina Nikoulina, Didier Schwab.
Findings of ACL 2021.
- 2020 **A Multilingual Neural Machine Translation Model for Biomedical Data**
Alexandre Bérard, Zae Myung Kim, Vassilina Nikoulina, Eunjeong Park, Matthias Gallé.
NLP-COVID19 at EMNLP 2020.
- 2020 **PATQUEST: Papago Translation Quality Estimation**
Yujin Baek*, Zae Myung Kim*, Jihyung Moon, Hyunjoong Kim, Eunjeong Park.
WMT 2020.

- 2017 **An Adaptive Vocabulary Learning Application through Modeling Learner's Linguistic Proficiency and Interests**
Zae Myung Kim, Suin Kim, Alice Oh, Ho-Jin Choi.
International Conference on Big Data and Smart Computing.
- 2016 **TIMEX3 and Event Extraction Using Recurrent Neural Networks**
Zae Myung Kim, Young-Seob Jeong, Ho-Jin Choi.
International Conference on Big Data and Smart Computing.
- 2016 **An Approach to Spam Comment Detection Through Domain-Independent Features**
Jong Myoung Kim, Zae Myung Kim, Kwangjo Kim.
International Conference on Big Data and Smart Computing.
- 2015 **Temporal Information Extraction from Korean Texts**
Young-Seob Jeong, Zae Myung Kim, Hyun-Woo Do, Chae-Gyun Lim, Ho-Jin Choi.
CoNLL 2015.
- 2017 **Modeling Long-Term Human Activeness Using Recurrent Neural Networks for Biometric Data**
Zae Myung Kim, Hyungrai Oh, Han-Gyu Kim, Chae-Gyun Lim, Kyo-Joong Oh, Ho-Jin Choi.
BMC Medical Informatics and Decision Making [SCIE, IF: 2.042].
- 2016 **Investigating the Impact of Possession-Way of a Smartphone on Action Recognition**
Zae Myung Kim*, Young-Seob Jeong*, Kyo-Joong Oh, Hyung-Rai Oh, Youssef Iraqi, Ho-Jin Choi.
Sensors 16(6) [SCIE, IF: 2.437].

Patent

- 2019 **Terminal, server and event suggesting methods thereof**
Hyungrai Oh, Ho-Jin Choi, Zae Myung Kim, Kyo-Joong Oh, Chae-Gyun Lim.
US Patent US10353975B2.

Teaching Experience

- Spring 2025 **Lecturer** – Natural Language Processing (CSCI 5541), University of Minnesota
[Apr 1, 2025] Lecture titled, *LLMs as Agents*.
- Fall 2024 **Head TA** – Intro. to Computer Programming in Java (CSCI 1103), University of Minnesota
- Fall 2024 **Lecturer** – Natural Language Processing (CSCI 5541), University of Minnesota
[Oct 31, 2024] Lecture titled, *LLMs as Agents*.
- Spring 2024 **TA, Lecturer** – Natural Language Processing (CSCI 5541), University of Minnesota
[Apr 2, 2024] Lecture titled, *Instructing and Augmenting LLMs*.
- Fall 2023 **TA, Lecturer** – Natural Language Processing (CSCI 5541), University of Minnesota
[Nov 21, 2023] Lecture titled, *Advanced Topics: Instructing and Augmenting LLMs*.
- Sept 2022 **Lecturer** – Elice Digital Transformation Training, Samsung Electronics
- Winter 2022 **TA** – Advanced Language Processing Winter School 2022
- Winter 2021 **TA** – Advanced Language Processing Winter School 2021
- Fall 2021 **Mentor** – Academic-Industrial Collaboration Project, Chungbuk National University
- Summer 2021 **Lecturer** – Three seminars on NLP trends and practical sessions, Rye & Catchers
- Fall 2016 **TA** – Elice AI & Data Science Course, Daejeon Creative Economy Innovation Center
- Fall 2015 **TA, Lecturer** – Introduction to AI (CS470), KAIST
[Sept 2, 2015] Lecture on extracting travel-related intents from textual review data.
- Summer 2015 **TA** – Machine Learning Summer Camp 2015 (Online), KAIST

Spring 2015 **TA** – Programming Practice (CS109), KAIST

Mentorship

Summer 2026 – Present **Shady Ali** (BS in Computer Science, University of Minnesota)

Summer 2025 – Spring 2026 **Yuxin Chen** (BS in Computer Science, University of Minnesota)

Spring 2025 **Yangfan He** (BS in Computer Science, University of Minnesota)

Spring 2024 – Spring 2025 **Robert Jia** (BS in Computer Science, University of Minnesota)
Now: Software Engineer at Microsoft

Summer 2024 – Spring 2025 **Bin Hu** (BS in Computer Science, University of Minnesota)
Now: MS in Computer Science, University of Maryland.

Spring 2024 – Spring 2025 **Ryan Koo** (MS in Computer Science, University of Minnesota)
Now: Software Engineer at Amazon

Fall 2023 – Spring 2024 **Preston Zhu** (BS in Computer Science, University of Minnesota)
Now: Software Engineer at Meta

Spring 2022 **Josh Spitzer-Resnick** (MS in Computer Science, University of Minnesota)
Now: Solventum Healthcare.

Invited Talks

Sept 21, 2026 **Seminar Talk** – Dept. of Artificial Intelligence, Sungkyunkwan University
Talk titled, *Metacognition for Self-Improving Agents: Who Judges the Judge, Who Improves the Improver*.

Jul 28, 2022 **Seminar Talk** – Dept. of Computer Engineering, Chungbuk Nat'l Univ.
Presented research on iterative text revision, machine translation, and discourse relations.

Dec 2, 2021 **Colloquium Talk** – Graduate School of Software, KAIST
Talk titled, *Cross-Lingual Discourse Analysis and Its Application*.

Oct 12, 2018 **Conference Talk** – DEVIEW 2018
Talk titled, *Papago Internals: Model Analysis and Development with Applied Technology*

Nov 11, 2016 **Seminar Talk** – Rye & Catchers
Talked about how textual data can be processed and utilized by *Bibly*, the company's book recommendation service.

Academic Services

2027 **AAAI 2027** (PC), **ICLR 2027** (Reviewer)

2026 **ICML 2026** (**Gold** Reviewer), **ICLR 2026** (Reviewer), **COLM 2026** (Reviewer)
NeurIPS 2026 (Reviewer), **EMNLP 2026** (Reviewer), **PLOS One** (Reviewer)
Expert Systems with Applications (Reviewer)

2025 **AAAI 2026** (PC), **ACL 2025** (Reviewer), **NeurIPS 2025** (Reviewer)

2024 **AAAI 2024** (PC), 7th **IWDS** (PC), **Knowledge-Based Systems** (Reviewer),
NAACL 2024 (Reviewer)

2023 **EMNLP 2023** (Reviewer), 6th **IWDS** (PC)

2022 5th **IWDS** (PC)

2021 4th **IWDS** (PC)

2020 **IEEE Access** (Reviewer), **NLP-COVID19 Workshop** at EMNLP 2020 (Reviewer)

2019 2nd **International Workshop on Dialog Systems (IWDS)** (PC)